# E6 Compound angle form A level only

Name: ____________________

Class: ____________________

Date: ____________________

Time: **352 minutes**

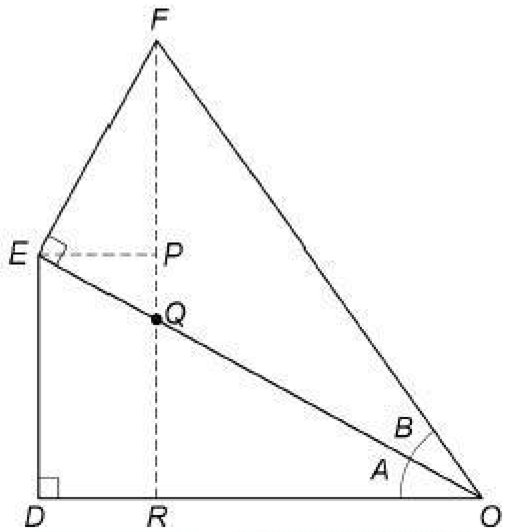
Marks: **295 marks**

Comments:

## Q1.

Some students are trying to prove an identity for sin($A$ + $B$).

They start by drawing two right-angled triangles $ODE$ and $OEF$, as shown.




The students' incomplete proof continues,

Let angle $DOE = A$ and angle $EOF = B$.

In triangle $OFR$,

Line 1 $\sin(A+B) = \dfrac{RF}{OF}$

Line 2 $= \dfrac{RP+PF}{OF}$

Line 3 $= \dfrac{DE}{OF} + \dfrac{PF}{OF}$ since $DE = RP$

Line 4 $= \dfrac{DE}{.....} \times \dfrac{.....}{OF} + \dfrac{PF}{EF} \times \dfrac{EF}{OF}$

Line 5 $= ............... + \cos A \sin B$

(a) Explain why $\dfrac{PF}{EF} \times \dfrac{EF}{OF}$ in Line 4 leads to cos$A$ sin$B$ in Line 5

**(2)**

(b) Complete Line 4 and Line 5 to prove the identity

Line 4 $= \dfrac{DE}{.....} \times \dfrac{.....}{OF} + \dfrac{PF}{EF} \times \dfrac{EF}{OF}$

Line 5 $= ............................................ + \cos A \sin B$

**(1)**

(c) Explain why the argument used in part **(a)** only proves the identity when *A* and *B* are acute angles.

**(1)**

(d) Another student claims that by replacing *B* with −*B* in the identity for sin(*A* + *B*) it is possible to find an identity for sin(*A* − *B*).

Assuming the identity for sin(*A* + *B*) is correct for all values of *A* and *B*, prove a similar result for sin(*A* − *B*).

**(3)**
**(Total 7 marks)**

## Q2.

(a) Determine a sequence of transformations which maps the graph of $y = \sin x$ onto the graph of $y = \sqrt{3}\sin x - 3\cos x + 4$

Fully justify your answer.

**(7)**

(b) (i) Show that the least value of $\dfrac{1}{\sqrt{3}\sin x - 3\cos x + 4}$ is $\dfrac{2-\sqrt{3}}{2}$

**(2)**

(ii) Find the greatest value of $\dfrac{1}{\sqrt{3}\sin x - 3\cos x + 4}$

**(1)**
**(Total 10 marks)**



## Q3.

(a) Prove the identity $\dfrac{\sin 2x}{1+\tan^2 x} \equiv 2\sin x\cos^3 x$

**(3)**

(b) Hence find $\displaystyle\int \frac{4\sin 4\theta}{1+\tan^2 2\theta}\,d\theta$

**(6)**
**(Total 9 marks)**

## Q4.

$f(x) = \sin x$

Using differentiation from first principles find the exact value of $f'\left(\dfrac{\pi}{6}\right)$
Fully justify your answer.

**(Total 6 marks)**

## Q5.

(a) Determine a sequence of transformations which maps the graph of $y = \cos\theta$ onto the graph of $y = 3\cos\theta + 3\sin\theta$

Fully justify your answer.

**(6)**

(b) Hence or otherwise find the least value and greatest value of

$$4 + (3\cos\theta + 3\sin\theta)^2$$

Fully justify your answer.

**(3)**
**(Total 9 marks)**

## Q6.

(a) Express $3 \cos x + 2 \sin x$ in the form R $\cos(x - \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, giving your value of $\alpha$ to the nearest 0.1°.

**(3)**

(ii) Hence find the minimum value of $3 \cos x + 2 \sin x$ and the value of $x$ in the interval
$0° < x < 360°$ where the minimum occurs. Give your value of $x$ to the nearest 0.1°.

**(3)**

(b) (i) Show that $\cot x - \sin 2x = \cot x \cos 2x$ for $0° < x < 180°$.

**(3)**

(ii) Hence, or otherwise, solve the equation

$$\cot x - \sin 2x = 0$$

in the interval $0° < x < 180°$.

**(3)**
**(Total 12 marks)**

## Q7.

The acute angles $\alpha$ and $\beta$ are given by $\tan \alpha = \frac{2}{\sqrt{5}}$ and $\tan \beta = \frac{1}{2}$.

(a) (i) Show that $\sin \alpha = \frac{2}{3}$, and find the exact value of $\cos \alpha$.

**(2)**

(ii) Hence find the exact value of sin $2\alpha$.

**(2)**

(b) Show that the exact value of $\cos(\alpha - \beta)$ can be expressed as $\frac{2}{15}(k + \sqrt{5})$, where $k$ is an integer.

**(4)**
**(Total 8 marks)**

## Q8.

The polynomial f($x$) is defined by $f(x) = 4x^3 - 11x - 3$.

(a) Use the Factor Theorem to show that $(2x + 3)$ is a factor of f($x$).

**(2)**

(b) Write f($x$) in the form $(2x + 3)(ax^2 + bx + c)$, where $a$, $b$ and $c$ are integers.

**(2)**

(c) (i) Show that the equation $2\cos 2\theta \sin\theta + 9\sin\theta + 3 = 0$ can be written as $4x^3 - 11x - 3 = 0$, where $x = \sin\theta$.

**(3)**

(ii) Hence find all solutions of the equation $2\cos 2\theta \sin\theta + 9\sin\theta + 3 = 0$ in the interval $0° < \theta < 360°$, giving your solutions to the nearest degree.

**(4)**
**(Total 11 marks)**

## Q9.

Angle $\alpha$ is acute and $\cos\alpha = \frac{3}{5}$. Angle $\beta$ is **obtuse** and $\sin\beta = \frac{1}{2}$.



(a) (i) Find the value of $\tan\alpha$ as a fraction.

**(1)**

(ii) Find the value of $\tan\beta$ in surd form.

**(2)**

(b) Hence show that $\tan(\alpha + \beta) = \frac{m\sqrt{3} - n}{n\sqrt{3} + m}$, where $m$ and $n$ are integers.

**(3)**
**(Total 6 marks)**

## Q10.

(a) Use the Factor Theorem to show that $4x - 3$ is a factor of

$$16x^3 + 11x - 15$$

**(2)**

(b) Given that $x = \cos\theta$, show that the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

can be written in the form

$$16x^3 + 11x - 15 = 0$$

**(4)**

(c) Hence show that the only solutions of the equation

$$27\cos\theta\cos 2\theta + 19\sin\theta\sin 2\theta - 15 = 0$$

are given by $\cos\theta = \frac{3}{4}$.

**(4)**

**(Total 10 marks)**

## Q11.

(a) Express $\sin x - 3\cos x$ in the form $R\sin(x - \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, giving your value of $\alpha$ to the nearest 0.1°.

**(3)**

(b) Hence find the values of $x$ in the interval $0° < x < 360°$ for which

$$\sin x - 3\cos x + 2 = 0$$

giving your values of $x$ to the nearest degree.

**(4)**

**(Total 7 marks)**



## Q12.

(a) Express $2\sin x + 5\cos x$ in the form $R\sin(x + \alpha)$, where $R > 0$ and $0° < \alpha < 90°$. Give your value of α to the nearest 0.1°.

**(3)**

(b) (i) Write down the maximum value of $2\sin x + 5\cos x$.

**(1)**

(ii) Find the value of $x$ in the interval $0° \le x \le 360°$ at which this maximum occurs, giving the value of $x$ to the nearest 0.1°.

**(2)**

**(Total 6 marks)**

## Q13.

(a) (i) Given that $\tan 2x + \tan x = 0$, show that $\tan x = 0$ or $\tan^2 x = 3$.

**(3)**

(ii) Hence find all solutions of $\tan 2x + \tan x = 0$ in the interval $0° < x < 180°$.

**(1)**

(b) (i) Given that $\cos x \neq 0$, show that the equation

$$\sin 2x = \cos x \cos 2x$$

can be written in the form

$$2 \sin^2 x + 2 \sin x - 1 = 0$$

**(3)**

(ii) Show that all solutions of the equation $2 \sin^2 x + 2 \sin x - 1 = 0$ are given by $\sin x = \dfrac{\sqrt{3}-1}{p}$, where $p$ is an integer.

**(3)**

**(Total 10 marks)**

## Q14.

(a) A curve is defined by the parametric equations $x = 3 \cos 2\theta$, $y = 2 \cos \theta$.

(i) Show that $\dfrac{dy}{dx} = \dfrac{1}{k\cos\theta}$, where $k$ is an integer.

**(4)**

(ii) Find an equation of the normal to the curve at the point where $\theta = \dfrac{\pi}{3}$.

**(4)**



(b) Find the exact value of $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sin^2 x \, dx$.



**(5)**

**(Total 13 marks)**

## Q15.

(a) Express $\cos x + 3 \sin x$ in the form $R \cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \dfrac{\pi}{2}$. Give your value of $\alpha$, in radians, to three decimal places.

**(3)**

(b) (i) Hence write down the minimum value of $\cos x + 3 \sin x$.

**(1)**

(ii) Find the value of $x$ in the interval $0 \leq x \leq 2\pi$ at which this minimum occurs, giving your answer, in radians, to three decimal places.

**(2)**

(c) Solve the equation $\cos x + 3 \sin x = 2$ in the interval $0 \le x \le 2\pi$, giving all solutions, in radians, to three decimal places.

**(4)**

**(Total 10 marks)**

## Q16.

(a) (i) Express $\sin 2\theta$ and $\cos 2\theta$ in terms of $\sin \theta$ and $\cos \theta$.

**(2)**

(ii) Given that $0 < \theta < \frac{\pi}{2}$ and $\cos \theta = \frac{3}{5}$, show that $\sin 2\theta = \frac{24}{25}$ and find the value of $\cos 2\theta$.

**(2)**

(b) A curve has parametric equations

$$x = 3 \sin 2\theta, \quad y = 4 \cos 2\theta$$

(i) Find $\frac{dy}{dx}$ in terms of $\theta$.

**(3)**

(ii) At the point $P$ on the curve, $\cos \theta = \frac{3}{5}$ and $0 < \theta < \frac{\pi}{2}$. Find an equation of the tangent to the curve at the point $P$.

**(3)**

**(Total 10 marks)**



## Q17.

(a) (i) Show that the equation $3 \cos 2x + 2 \sin x + 1 = 0$ can be written in the form

$$3 \sin^2 x - \sin x - 2 = 0$$

**(3)**

(ii) Hence, given that $3 \cos 2x + 2 \sin x + 1 = 0$, find the possible values of $\sin x$.

**(2)**

(b) (i) Express $3 \cos 2x + 2 \sin 2x$ in the form $R \cos (2x - \alpha)$, where $R > 0$ and $0° < \alpha < 90°$, giving $\alpha$ to the nearest 0.1°.

**(3)**

(ii) Hence solve the equation

$$3 \cos 2x + 2 \sin 2x + 1 = 0$$

for all solutions in the interval $0° < x < 180°$, giving $x$ to the nearest 0.1°.

**(3)**
**(Total 11 marks)**

## Q18.

(a) Express $\sin x - 3\cos x$ in the form $R\sin(x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.
Give your value of $\alpha$ in radians to two decimal places.

**(3)**

(b) Hence:

(i) write down the minimum value of $\sin x - 3\cos x$;

**(1)**

(ii) find the value of $x$ in the interval $0 < x < 2\pi$ at which this minimum value occurs, giving your value of $x$ in radians to two decimal places.

**(2)**
**(Total 6 marks)**

## Q19.

(a) Express $\sin 2x$ in terms of $\sin x$ and $\cos x$.

**(1)**

(b) Solve the equation

$$5\sin 2x + 3\cos x = 0$$

giving all solutions in the interval $0° \le x \le 360°$ to the nearest 0.1°, where appropriate.



**(4)**

(c) Given that $\sin 2x + \cos 2x = 1 + \sin x$ and $\sin x \neq 0$, show that $2(\cos x - \sin x) = 1$.

**(4)**
**(Total 9 marks)**

## Q20.

(a) (i) Show that the equation $3\cos 2x + 7\cos x + 5 = 0$ can be written in the form $a\cos^2 x + b\cos x + c = 0$, where $a$, $b$ and $c$ are integers.

**(3)**

(ii) Hence find the possible values of $\cos x$.

**(2)**

(b) (i) Express $7\sin\theta + 3\cos\theta$ in the form $R\sin(\theta + \alpha)$, where $R > 0$ and $\alpha$ is an acute angle. Give your value of $\alpha$ to the nearest 0.1°.

**(3)**

(ii) Hence solve the equation 7 sin $\theta$ + 3 cos $\theta$ = 4 for all solutions in the interval 0° ≤ $\theta$ ≤ 360°, giving $\theta$ to the nearest 0.1°.

**(3)**

(c) (i) Given that $\beta$ is an acute angle and that $\tan\beta = 2\sqrt{2}$, show that $\cos\beta = \frac{1}{3}$.

**(2)**

(ii) Hence show that $\sin 2\beta = p\sqrt{2}$, where $p$ is a rational number.

**(2)**

**(Total 15 marks)**

## Q21.

(a) (i) Express 6 sin $\theta$ + 8 cos $\theta$ in the form $R$ sin($\theta$ + $\alpha$), where $R$ > 0 and 0° < $\alpha$ < 90°. Give your value for $\alpha$ to the nearest 0.1°.

**(2)**

(ii) Hence solve the equation 6 sin $2x$ + 8 cos $2x$ = 7, giving all solutions to the nearest 0.1° in the interval 0°< $x$ < 360°.

**(4)**

(b) (i) Prove the identity $\frac{\sin 2x}{1-\cos 2x} = \frac{1}{\tan x}$.

**(4)**

(ii) Hence solve the equation

$$\frac{\sin 2x}{1-\cos 2x} = \tan x$$



giving all solutions in the interval 0° < $x$ < 360°.

**(4)**

**(Total 14 marks)**

## Q22.

(a) By writing sin $3x$ as sin($x$ + $2x$), show that sin $3x$ = 3 sin $x$ – 4 $\sin^3 x$ for all values of $x$.

**(5)**

(b) Hence, or otherwise, find $\int \sin^3 x\,dx$.

**(3)**

**(Total 8 marks)**

## Q23.

(a) The angle $\alpha$ is acute and $\sin \alpha = \frac{4}{5}$.

(i) Find the value of $\cos \alpha$.

**(1)**

(ii) Express $\cos(\alpha - \beta)$ in terms of $\sin \beta$ and $\cos \beta$.

**(2)**

(iii) Given also that the angle $\beta$ is acute and $\cos \beta = \frac{5}{13}$, find the exact value of $\cos(\alpha - \beta)$.

**(2)**

(b) (i) Given that $\tan 2x = 1$, show that $\tan^2 x + 2 \tan x - 1 = 0$.

**(2)**

(ii) Hence, given that $\tan 45° = 1$, show that $\tan 22\frac{1}{2}° = \sqrt{2} - 1$

**(3)**

**(Total 10 marks)**

## Q24.

(a) Express $\cos 2x$ in terms of $\sin x$.

**(1)**

(b) (i) Hence show that $3 \sin x - \cos 2x = 2 \sin^2 x + 3 \sin x - 1$ for all values of $x$.

**(2)**



(ii) Solve the equation $3 \sin x - \cos 2x = 1$ for $0° < x < 360°$.



**(4)**

(c) Use your answer from part (a) to find $\int \sin^2 x \, dx$.

**(2)**

**(Total 9 marks)**

## Q25.

(a) Use the identity

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

to express $\tan 2x$ in terms of $\tan x$.

**(2)**

(b) Show that

$$2-2\tan x-\frac{2\tan x}{\tan 2x}=(1-\tan x)^2$$

for all values of $x$, $\tan 2x \neq 0$.

**(4)**
**(Total 6 marks)**

## Q26.

(a) Express $4\cos x + 3\sin x$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0° < \alpha < 360°$, giving your value for $\alpha$ to the nearest $0.1°$.

**(3)**

(b) Hence solve the equation $4\cos x + 3\sin x = 2$ in the interval $0° < x < 360°$, giving all solutions to the nearest $0.1°$.

**(4)**

(c) Write down the minimum value of $4\cos x + 3\sin x$ and find the value of $x$ in the interval $0° < x < 360°$ at which this minimum value occurs.

**(3)**
**(Total 10 marks)**

## Q27.

It is given that $3\cos\theta - 2\sin\theta = R\cos(\theta + \alpha)$, where $R > 0$ and $0° < \alpha < 90°$.

(a) Find the value of $R$.

**(1)**

(b) Show that $\alpha \approx 33.7°$.

**(2)**



(c) Hence write down the maximum value of $3\cos\theta - 2\sin\theta$ and find a **positive** value of $\theta$ at which this maximum value occurs.

**(3)**
**(Total 6 marks)**

## Q28.

(a) Express $\cos 2x$ in the form $a\cos^2 x + b$, where $a$ and $b$ are constants.

**(2)**

(b) Hence show that $\int_0^{\frac{\pi}{2}} \cos^2 x\, dx = \frac{\pi}{a}$, where $a$ is an integer.

**(5)**
**(Total 7 marks)**

**Q29.**

(a) (i) Express sin $2x$ in terms of sin $x$ and cos $x$.

**(1)**

(ii) Express cos $2x$ in terms of cos $x$.

**(1)**

(b) Show that

sin $2x$ – tan $x$ = tan $x$ cos $2x$

for all values of $x$.

**(3)**

(c) Solve the equation sin $2x$ – tan $x$ = 0, giving all solutions in degrees in the interval $0° < x < 360°$.

**(4)**

**(Total 9 marks)**

**Q30.**

(a) Express 2 sin $x$ + cos $x$ in the form $R$ sin($x + \alpha$) where $R$ is a positive constant and α is an acute angle. Give your value of $\alpha$ to the nearest 0.1°.

**(3)**

(b) Solve the equation 2 sin$x$ + cos $x$ = 1 for $0° \le x < 360°$.

**(4)**

**(Total 7 marks)**

**Q31.**



(a) Express sin $2x$ in terms of sin $x$ and cos $x$.



**(1)**

(b) Using the identity $\cos(A + B) = \cos A \cos B - \sin A \sin B$ :

(i) express cos $2x$ in terms of sin $x$ and cos $x$;

**(2)**

(ii) show, by writing $3x$ as $(2x + x)$, that

$$\cos 3x = 4\cos^3 x - 3\cos x$$

**(4)**

(c) Show that $\int_0^{\frac{\pi}{2}} \cos^3 x \, dx = \frac{2}{3}$.

**(5)**

**(Total 12 marks)**

**Q32.**

A particle is projected from a horizontal surface and lands on the same horizontal surface. Its initial velocity is $V$ m s$^{-1}$ at an angle $\alpha$ above the horizontal.

(a) Prove that the range of the particle is

$$\frac{V^2 \sin 2\alpha}{g}.$$

**(6)**

(b) Hence explain why the maximum range of the particle is $\frac{V^2}{g}$, and state the value of $\alpha$ for which the particle has this maximum range.

**(2)**

(c) A machine for launching tennis balls fires the balls from ground level on a horizontal surface. Assume that the balls are all fired at the same speed. The maximum range of one of these balls is 35 metres.

(i) Find the speed at which the tennis balls are fired.

**(2)**

(ii) Find the range of a tennis ball fired at an angle of 30° above the horizontal, giving your answer to the nearest metre.

**(2)**

**(Total 12 marks)**